Growth and Maturation in Athletic Development

Purpose of this document

This document was inspired by a Top Up Tuesday delivered by Gill Myburgh, whose work focuses on managing youth academies within the English Premier League system.

Its purpose is to help coaches, sport scientists, and practitioners accurately interpret what they are seeing when athletes move through puberty, and to make **better, more informed decisions**, not softer ones.

Growth and maturation are not excuses for reduced standards, they are merely **contexts that shape capacity, tolerance, and adaptation**. When that context is understood, load, risk, expectations, and long-term development can be managed deliberately.

Design principle: Biological age > chronological age. Always.

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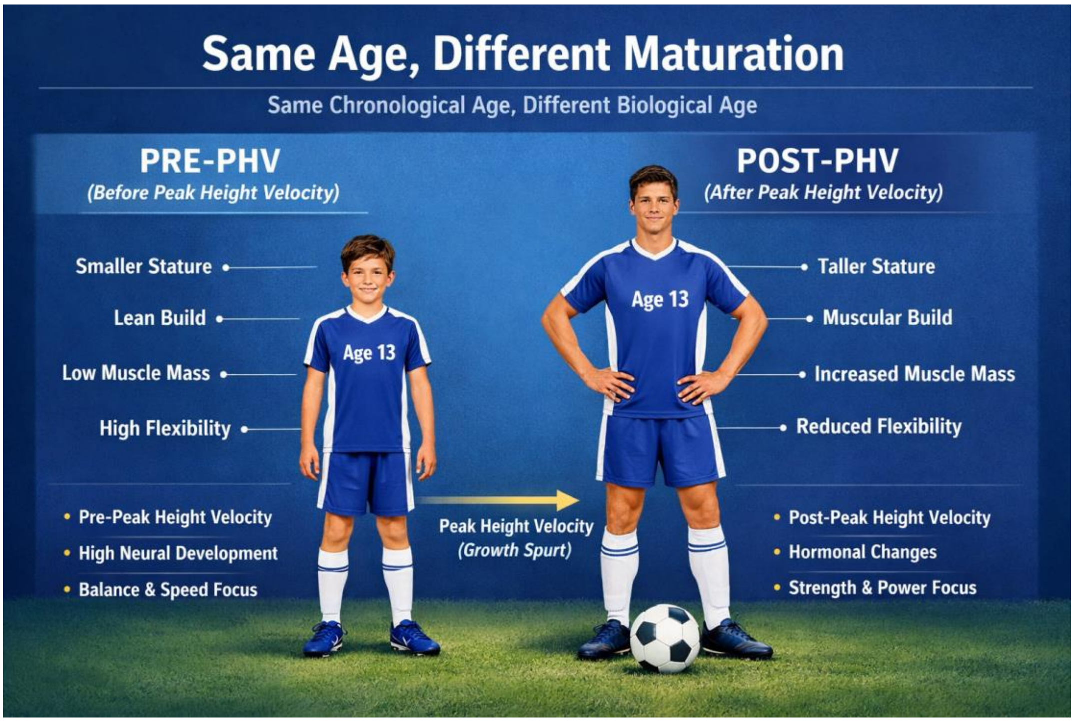
1. Chronological Age vs Biological Age

Chronological age tells you *how old the athlete is*. Biological age tells you *what the athlete can currently tolerate and adapt to*. Training programmes that ignore this difference are not evidence-based, they're convenient.

Key idea: Two athletes can be the same age and be in completely different developmental states. Treating them the same is poor player management.



Training implications by maturation stage

- **Pre-adolescents:** respond best to high neural-demand activities (sprinting, jumping, coordination).
- **Adolescents:** respond better to combined neural + structural stimuli (strength training + plyometrics).
- **Circa PHV (During Growth Spurt):** balance, coordination, and movement control temporarily decline, this is not laziness or poor focus.

Analogy: You wouldn't tune an engine while the chassis is being stretched.

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2. Hormonal Environment & Adaptation

Adolescents respond more favourably to hypertrophy-based training than pre-adolescents due to increased concentrations of **testosterone** and **growth hormone**.

This is where the phrase holds: **"First you stretch them out, then you fill them out."**

Trying to force size gains before the hormonal environment is ready is wasted effort and unnecessary stress.

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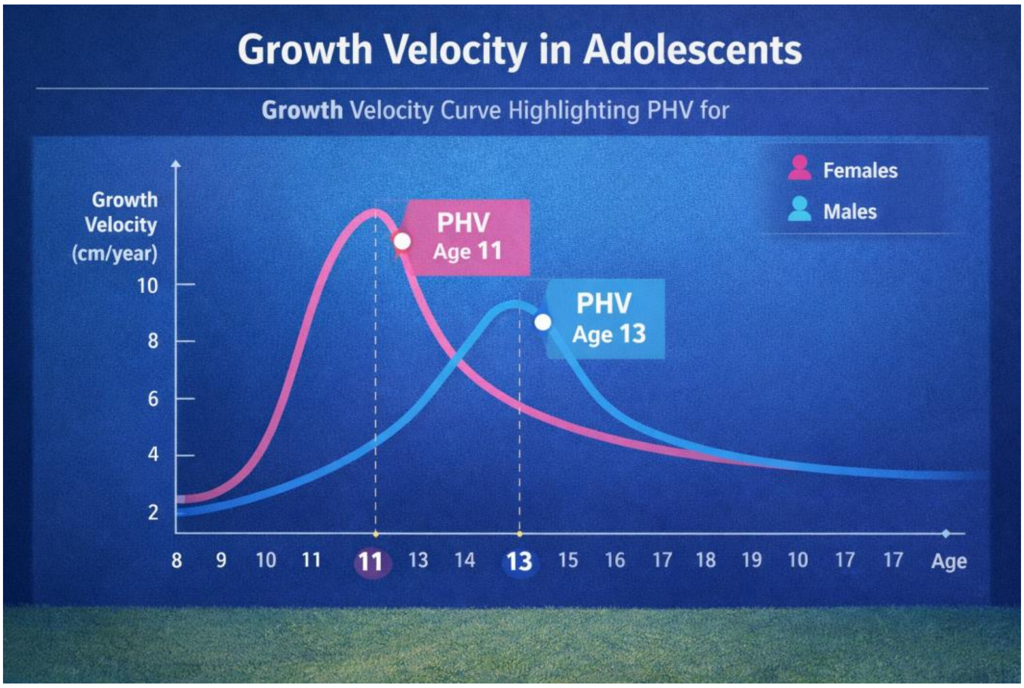
3. Peak Height Velocity (PHV)

Definition:
Peak Height Velocity (PHV) is the period during adolescence where an athlete experiences their fastest rate of growth in stature.

- Females: ±11 years
- Males: ±13 years
- Males generally experience a **larger magnitude** of growth

PHV is not a date on a calendar. It's a *moving target*.



Why PHV matters

During PHV:

- Limb lengths change faster than neuromuscular control can adapt
- Coordination temporarily decreases
- Trunk stability often declines (especially in boys)
- Injury risk increases if load is poorly managed

Ignoring PHV is how you lose athletes, not how you build them.

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4. Monitoring Growth & Maturation

You don't manage what you don't monitor. This doesn't require expensive tech, it requires consistency.

Methods of predicting PHV

Invasive methods (not recommended for routine use)

- **FELS hand–wrist X-ray**
  - Radiographic exposure
  - Time-intensive
- **Tanner Staging**
  - Genital/breast development assessment
  - Highly invasive and context-sensitive

Non-invasive methods (preferred)

- **Mirwald equation** (sitting height, leg length, body mass)

1. Population Bias (Big One)

Mirwald was developed on:

Mostly **white, Western populations**

Relatively homogeneous growth patterns

Stable nutrition and health environments

On the African continent, you often have:

Greater **genetic diversity**

Wider variation in **growth tempo**

More variability in **early-life nutrition**

Different limb-to-trunk proportions in some populations

Mirwald Equation Accuracy **(Under Perfect Conditions) :** In the original validation samples (mainly European/North American youth):

- **Standard error ±0.5–1.0 years**
- Accuracy is best **around PHV**
- Accuracy drops **before and after PHV**

That means even in perfect conditions, you can be **off by up to a year**. Therefore, on the African continent the Standard Error is even greater.

- **Khamis–Roche method** (% predicted adult height) On average:

  Error ≈ ±2–3 cm for predicted adult height.

  Roughly 90–95% of predictions fall within ±5 cm.

Recommendation: Assess growth **2–3 times per year**.

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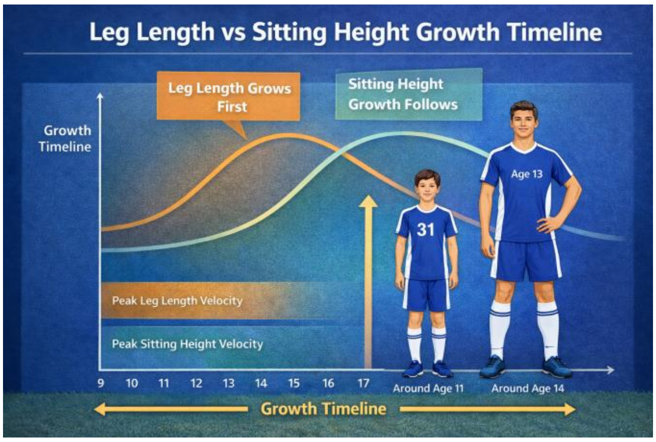
5. Growth Velocity: What You're Actually Seeing

Growth does not happen evenly.

- **Leg length increases first**
- **Sitting height follows**

If a 13-year-old male has a predicted PHV at 13.4, you don't wait until problems appear, you prepare.



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6. PHV vs PWV (Peak Weight Velocity)

- **PHV comes first** (rapid height gain)
- **PWV follows** (mass and muscle accumulation)

Understanding where the athlete sits on this spectrum matters more than age.

- PHV-dominant athlete → coordination risk, loading sensitivity
- PWV-dominant athlete → greater strength tolerance

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7. Importance for Athletic Development

Historically, these periods were called **"windows of opportunity."** That language is misleading and harmful.

It implies:

- Opportunities can be missed forever
- Athletic ceilings are fixed

Current thinking uses:

Periods of Accelerated Adaptation

This means:

- Adaptation is always possible
- Some periods simply allow *faster* gains

Training implications:

- Load tolerance increases
- Frequency can increase
- Larger loading units may be handled (*if* managed correctly).

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8. Training Considerations & Risk Profiling

This system should function as a **risk flagging model**, not a restriction model.

Goal: Keep the athlete playing *as long as possible*.

Key realities

- Only 25% of athletes experience significant adolescent awkwardness
- Growth pain ≠ injury
- Removal from training is a last resort, not a default

Common conditions to monitor:

- Osgood–Schlatter disease
- Sever's disease

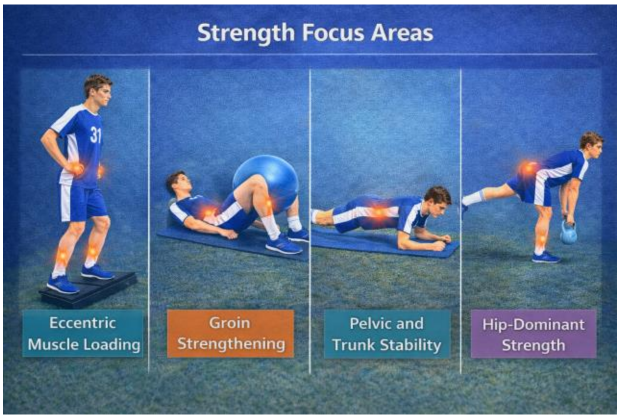
Training Priorities During the Peak Height Velocity Phase

- Balance & coordination challenges (within pain tolerance)
- Communication with coaches (e.g. modified warm-ups)
- Maintain exposure where possible

Strength focus areas

- Eccentric muscle loading
- Groin strengthening
- Pelvic and trunk stability
- Hip-dominant strength

Trunk stability loss during PHV is expected. Don't neglect this part.



Final Takeaway

If you don't understand growth and maturation:

- You'll mislabel effort as attitude
- You'll mistake coordination loss for regression
- You'll either overload or under-stimulate

If you *do* understand it:

- You keep athletes healthier
- You develop them faster
- You earn trust without lowering standards

That's the job.